

Pediatric and Adolescent Hip Injuries

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Almost a decade ago it was estimated that over 30 million children participated in organized sports programs, with a third of them suffering an injury yearly that required evaluation by a nurse or physician.¹ There are physical and physiologic differences between the adolescent and adult athlete that may cause the younger athlete to be more vulnerable to injury. Children may not have as much coordination and musculoskeletal balance, as limb mass increases faster than limb length.⁶¹ Muscle tendon growth lags behind bone growth and growing cartilage is more susceptible to stress and injury. These sports injuries are partly due to children's increased participation in sports and the propensity for children to focus on a single sport, leading to an increased risk of overuse injuries. More recently it has been reported that up to 2.6 million young athletes each year have injuries that warrant a visit to the emergency room, costing almost \$2 billion yearly.⁵⁵

Injuries to the hip and pelvis in pediatric and adolescent patients have been reported in the past several decades,^{68,82,90,91,144,179} but are now receiving increasing interest because of advances in arthroscopic treatment and magnetic resonance imaging (MRI). Injuries around the hip can be caused by a single traumatic event or repetitive microtrauma. The majority of injuries around the hip in children are soft tissue, apophyseal, or bony injuries that require supportive management, with the minority of injuries requiring surgical intervention. There is likely an effect of sex and age on hip injuries due to sports, with females sustaining more chronic overuse and soft tissue injuries compared with males.¹⁷¹ This chapter offers an overview of common injuries around the hip in the pediatric and adolescent athlete.

BONY INJURIES

Avulsion Fractures

Apophyseal avulsion fractures of the pelvis are injuries that occur almost exclusively in adolescence. For example, a sudden, violent, unbalanced muscle contraction is applied through the musculotendinous units during sporting activities. The cartilaginous growth plate at the apophysis fails, resulting in an avulsion fracture of the pelvis.¹¹⁸ Once the injury has occurred, the bony displacement is limited by the periosteum and the surrounding muscle fascia. Sports such as soccer, rugby, ice hockey, gymnastics, and sprinting that involve kicking, rapid acceleration and deceleration, and jumping are commonly associated with avulsion fractures. Intensive training exposes the epiphyseal plate to repeated

tensile stress while overstrengthening the muscles around the joint. The inherent weakness of the epiphyseal plate combined with the increased demand on the musculature predispose the adolescent athlete to avulsion injury.

The sites for apophyseal injury around the hip include the ischium (hamstring attachment), anteroinferior iliac spine (AIIS; rectus femoris), anterosuperior iliac spine (ASIS; sartorius), iliac crest (abdominal musculature), lesser trochanter (iliopsoas) (Fig. 136.1), and greater trochanter (abductors). In the two largest studies of avulsion fractures, the most common location of such fractures were the AIIS in 33%, ASIS in 28%, and the ischial tuberosity (IT) in 30%, affecting males predominantly (75%).^{153,158} The clinical presentation typically follows a traumatic incident with an acute onset of localized pain and the description of a "pop." Palpation and passive stretching of the muscle is typically quite painful, and patients will assume a position that places the least tension on the involved muscle. Although clinical presentation is often diagnostic, radiographic imaging is useful to determine the location, size, and degree of bony displacement.

Initial management of the majority of avulsion fractures is usually conservative, including rest and ice, followed by protected weight bearing with crutches until symptoms resolve.⁷¹ Light isometric stretching, full weight bearing, and strengthening exercises are started when the patient is free of pain. Return to sports occurs after full pain-free range of motion (ROM) and full strength are achieved, which can take anywhere from 8 weeks to 6 months. A recent meta-analysis found that patients undergoing surgery had a higher overall success rate and return to sports compared with those treated conservatively.⁴¹ However, there is usually a need for surgical intervention in avulsion fractures where the fragment has displaced more than 1.5 cm and there is loss of function or painful nonunion. In the case of the IT avulsion fracture, several reports have advocated repair if the bony fragment is greater than 2 cm; otherwise the fibrous nonunion can lead to chronic buttock pain, decreased hamstring strength, or potentially symptoms affecting the sciatic nerve.^{31,54,71,159,184} Although it is generally agreed that large displaced fragments (>2 cm) may require surgical fixation, the optimal timing of surgery still remains unclear (Fig. 136.2).

Stress Fractures

Stress fractures of the femoral neck are most commonly seen in distance runners and young people enlisted in the military; the incidence is as high as 21% to 31% among these groups.^{13,51,122}

Similarly, the incidence of stress fractures of the femoral neck in young female athletes has been reported to be up to 20%.²² Such fractures can also occur in children younger than 10 years of age.¹⁹ Stress fractures originate from either abnormal forces on normal bone (fatigue fractures) or normal forces on abnormal bone (insufficiency fractures).¹⁴⁵ These overuse injuries result from repetitive microtrauma on the bone and occur when the extent of damage exceeds the remodeling process. The most common sites of stress fractures in the hip area are in the femur, pubic rami, iliac crest, and sacroiliac joints.¹¹⁰ Stress fractures of the femoral neck are particularly important because they often go untreated or unrecognized and can lead to the potentially devastating consequence of an acute femoral neck fracture.^{83,143} Female athletes are more likely to develop stress fractures because of the classic female athlete triad: amenorrhea, eating disorders, and low bone density.^{59,143,151} Additional risk factors

include chronic glucocorticoid use, smoking, hyperparathyroidism, hyperthyroidism, malabsorption syndromes, and calcium deficiencies.^{14,34,36,48,57,60,117,125,127,129,134,178}

Stress fractures of the hip often present with only subtle clinical symptoms, making the diagnosis difficult. Hip pain associated with these stress fractures presents with an insidious onset of vague groin discomfort that is worse with activity and alleviated by rest. Patients often provide a history of a recent increase in their running or activity. As the fracture worsens, pain may occur earlier or even at rest. On physical examination, pain may be reproduced at the extremes of ROM, particularly with internal rotation.²¹ A positive Trendelenburg test, inability to do a straight leg raise against resistance, and inability to hop on the affected side should raise suspicion of a stress fracture.¹³⁷ Plain radiographs are often negative in the acute setting; therefore advanced imaging may be required. Bone scan or MRI can be more helpful for diagnosing a stress fracture and should be considered if plain films are not diagnostic (Fig. 136.3).

The management of hip stress fractures is based on location, chronicity, and causative factors.¹⁴⁴ Femoral neck fractures occur in two different patterns, tension versus compression.³⁸ Tension-sided stress fractures are located on the superolateral femoral neck, more commonly in adults, and they may become displaced. Management of these types of fractures should be with percutaneous cannulated screws placed up the femoral neck. Tension-sided compression fractures are at higher risk for non-union, deformity, and avascular necrosis. Compression-sided fractures occur in the inferomedial neck; they rarely displace but can cause a mild varus deformity. Younger patients typically present with compression-sided femoral neck stress fractures, which can be managed conservatively.^{107,168} In most adolescents, conservative treatment includes non-weight bearing on the affected leg until radiographic union, nonsteroidal antiinflammatory drugs (NSAIDs), gentle ROM exercises, and regional muscular strengthening can be initiated. If healing does not occur within 6 to 8 weeks, use of a bone stimulator may be tried. Return to play should be allowed only after complete healing on clinical and radiographic examination. Treatment of any



Fig. 136.1 Avulsion fracture of the lesser tuberosity in a skeletally immature patient. The white arrow denotes the site of avulsion.

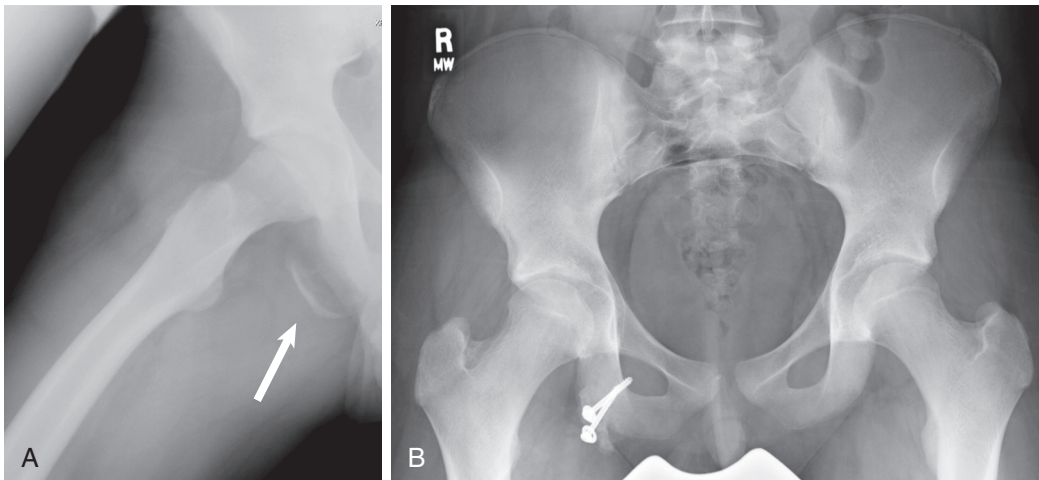


Fig. 136.2 (A) Avulsion fracture of the ischial tuberosity. The white arrow denotes a large fragment of ischium. (B) The same patient after open reduction and internal fixation



Fig. 136.3 Magnetic resonance image of a stress fracture of the intertrochanteric region of the hip in an adolescent female. The white arrow denotes an area of edema due to the stress fracture.

underlying endocrine or other disorder is paramount to prevent recurrence.

Fractures

Fractures around the hip are relatively rare in the adolescent and pediatric population, accounting for less than 1% of fractures. These fractures usually result from high-energy trauma or a fall from a height rather than from participation in sports. Fractures can occur in the femoral head and femoral neck, the subtrochanteric region of the femur, and around the pelvic ring. Fractures around the pelvis can occur as single breaks due to the elasticity of the child's pelvis and can injure the triradiate cartilage, leading to growth disturbances of the acetabulum.^{27,63,64} Radiographs of the pelvis and orthogonal views of the hip are usually diagnostic, but computed tomography (CT) can be useful in many cases.

Fractures of the acetabulum are generally treated nonoperatively in cases of minimal displacement, disruption of a small fragment, stable fracture patterns, or Salter-Harris I or II fractures of the triradiate. Operative intervention has been recommended in comminuted, open, and unstable fracture patterns; however, long-term results have not been encouraging.^{27,62–64}

Fractures of the femoral head and neck are classified by the method of Delbet, comprising type I (transphyseal), type II (transcervical), type III (cervicotrochanteric), and type IV (intertrochanteric) fractures.³² Type I fractures are the least common but have the highest rate of osteonecrosis (38%–100%, depending on the presence of femoral head dislocation).^{126,144} Irreducible fractures require open reduction and internal fixation.¹⁶⁰ Types II and III should be treated with anatomic reduction and internal fixation¹⁶⁰; they are the most commonly encountered femoral neck fracture in children. Type II fractures have approximately twice the risk of avascular necrosis of type III fractures.^{126,144} There may be some role for decompression of the capsular hematoma associated with femoral neck fractures, as lower rates of avascular necrosis were seen in patients who had a capsular decompression.¹³⁶ Type IV fractures and subtrochanteric femoral

fractures are extracapsular and have the most favorable prognosis. These fractures should be treated with anatomic reduction and internal fixation with blade plate fixation or screw side plate fixation.^{2,18,84,106} In some cases intramedullary nailing or a dynamic hip screw can be used for subtrochanteric femoral fractures.¹⁶⁹ Postoperatively these patients are generally instructed to ambulate with crutches and protected weight bearing.

Dislocation

Hip dislocation is relatively uncommon in the adolescent population, accounting for less than 5% of all traumatic hip dislocations. Traumatic hip dislocation in children differs from the same in adults in that it is classified as a low-energy injury in the majority of patients when it results from a sports injury or fall from a relatively low height.^{116,159,176,189} This difference is due to increased joint laxity in children, making dislocation possible with relatively minor trauma. Proximal femoral deformity has also been postulated to cause instability and may contribute to posterior hip dislocation.^{109,138}

Although the femoral head in children may dislocate in any direction, posterior dislocations occur more frequently, accounting for over 90% of all hip dislocations.^{10,93,116,176} The hip is usually held in flexion, adduction, and internal rotation with a posterior dislocation. The neurovascular status must be documented thoroughly and blunt trauma to the ipsilateral knee should be noted. Diagnosis can be made by history and physical examination. The femoral head can sometimes be palpated posteriorly. Radiographs, which should include a view of the entire pelvis and orthogonal views of the affected hip, are usually diagnostic. Radiographs should be examined carefully in order to evaluate for ipsilateral physeal injury or an associated femoral neck fracture. A CT scan can be useful to detect bony fragments within the joint as well as concomitant injury; however, MRI may be coming the modality of choice.¹¹¹

The treatment for a traumatic hip dislocation is an emergent closed reduction within the first 6 hours of the time of injury so as to minimize the risk of avascular necrosis. The incidence of avascular necrosis is about 5% to 15% of all patients, but it increases by as much as 20-fold if the time to reduction is after 6 hours.^{93,116} Closed reduction can be performed in the emergency room, but we prefer reduction in the operating room with general anesthesia and imaging (Fig. 136.4). Open reduction is performed only if the hip is irreducible after two to three reduction attempts or if soft tissue or bony fragments are interposed between the head and the acetabulum. The approach should be the same as the direction of the dislocation—that is, a posterior approach for a posterior dislocation. The acetabulum should be cleared of any debris, osteochondral fragments fixed if large enough, and the soft tissue repaired if possible.

The major complication after a dislocated hip is avascular necrosis of the femoral head.^{11,116,141,176} When a traumatic dislocation is associated with femoral epiphysiolysis, the risk of avascular necrosis is almost 100%.^{10,66,86,140} A patient who has had a delayed reduction should undergo MRI to assess for avascular necrosis 3 to 6 months after the injury. An alternative algorithm has been proposed to obtain an MRI 6 weeks after injury. If there is normal marrow signal, no further imaging is necessary; if abnormal

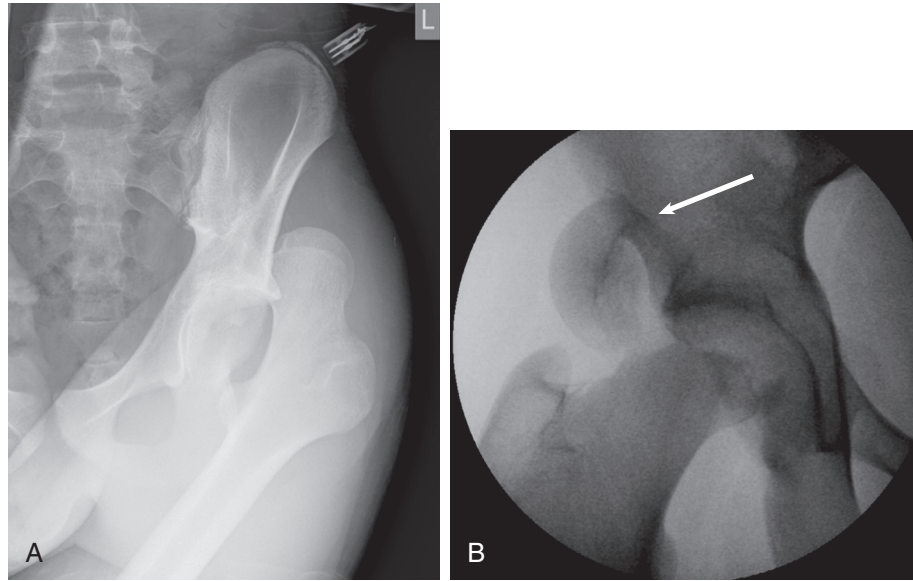


Fig. 136.4 (A) Posterior dislocation of the hip in a skeletally immature patient. (B) Epiphysiolsis during reduction of a dislocated hip. Open reduction and internal fixation of the epiphysis was necessary

signal is noted, a follow-up scan should be performed at 3 months.¹⁵⁰ Other complications include sciatic nerve injury, late posttraumatic osteoarthritis, coxa magna, heterotopic ossification, and recurrent dislocation.¹⁸⁹

There have been some studies on intra-articular pathology evaluation by hip arthroscopy following hip dislocation,^{77,147} which have been corroborated by an open surgical approach.¹³⁸ In all patients, there was documentation of intra-articular pathology, including chondral damage, labral tears, ligamentum teres tears, and adhesions.¹⁵⁴ Hip arthroscopy in the acute setting may be difficult because of fluid extravasation and should be considered only in the most experienced hands.

Pathologic Lesions

Very rarely both benign and malignant lesions may be found when the patient with hip pain is being evaluated. The most common benign lesions are simple bone cysts and osteoid osteomas, whereas the most common malignant tumors are osteosarcoma and Ewing sarcoma.¹⁵⁵ Radiographs and advanced imaging (CT and/or MRI) should be obtained to evaluate the lesion's location and size and to check for the presence of fracture as well as soft tissue extension and metastatic manifestations. Some simple benign lesions heal or regress spontaneously and need only close observation, whereas others, if quite large, may need surgical intervention. If malignancy is suspected, referral to orthopaedic oncology is advisable, as adjunctive and wide resection surgery may be indicated.

SOFT TISSUE INJURIES

Muscle Injury

Muscle strains and contusions are common in the young athlete.⁴⁹ Muscle strains are associated with improper warm-up, previous injury, fatigue, weather conditions, and uneven surfaces among

other factors.^{15,16} A muscle strain occurs in the hip area when muscle is functioning eccentrically—that is, the muscle is contracting while being lengthened. The most susceptible muscles involved are those that cross two joints, in particular the rectus femoris, hamstrings, and adductors.¹⁶ A fall or direct blow causes a muscle contusion, which is commonly seen in contact sports.

The diagnosis of a muscle strain or contusion is typically straightforward. For a muscle strain, a high-velocity activity is usually being performed and the patient reports a sudden and intense pain in the affected muscle. There is usually localized pain and tenderness of the affected muscle with varying degrees of ecchymosis. The patient is usually unable to continue the activity and to use the muscle immediately. A contusion can be distinguished from a muscle rupture by the fact that with a contusion or strain the ability to continue using the muscle remains; the distinction can also be made by the mechanism of injury. Nonsurgical treatment varies, but all approaches involve the same principles. An early compression wrap and icing will help to control edema, bleeding, and swelling. NSAIDs are used for the first several days. Immobilization and partial weight bearing can help to provide comfort and muscle protection in the early period. Early ROM should comprise gentle stretching with progression to strengthening when muscle pain has resolved. Functional activities can start once strength has returned. Surgical intervention is rarely indicated.

Myositis ossificans can occur after moderate or severe injuries.¹⁵ The vastus intermedius of the quadriceps is most commonly affected in this condition. It is usually diagnosed clinically by persistent soft tissue swelling, particularly with a muscle that is warm and tender to the touch. A radiograph will become evident, with heterotopic bone formation around 2 to 4 weeks after the injury. The size of the myositis ossificans usually stabilizes by 6 months but can cause continued pain and may delay rehabilitation.¹⁸⁰ Surgery is usually not indicated.

Bursitis

Bursitis occurs at two major areas around the hip, the iliopsoas bursa and the greater trochanteric bursa. The iliopsoas bursa is the largest synovial bursa in the body and is located between the iliopsoas tendon and the lesser trochanter, extending upward into the iliac fossa beneath the iliacus muscle.²⁴ The greater trochanteric bursa is located on the lateral side of the greater trochanter and cushions the gluteal tendons, iliotibial band, and tensor fascia latae.¹⁶¹ A smaller, less common bursa is the iliopectineal bursa, which is situated between the iliopsoas muscle belly and the femoral head. Causes of bursitis include chronic microtrauma, arthritis, regional muscle dysfunction, overuse, and acute injury.^{56,161} Hip bursitis is commonly caused by repetitive motion of the hip during cycling or running or a direct injury to the hip such as by a fall or a tackle onto the greater trochanter.

Clinical presentation depends on which bursa is affected. Iliopsoas bursitis causes pain the anterior groin, tenderness to palpation over the lesser trochanter, and pain with resisted hip flexion (Stinchfield test).¹¹⁴ Symptoms of trochanteric bursitis include pain on the lateral aspect of the hip and thigh, pain from sitting to standing, and pain on walking up stairs. Having the patient lie on his or her side and reproduce the motion of riding a bike can elicit tenderness. In both cases, erythema, swelling, and warmth may be present over the front or side of the hip. Diagnosis is usually clinical and imaging is often unnecessary. Plain films can sometimes show round calcifications that are isolated around the greater trochanter in the case of trochanteric bursitis.¹⁶¹ Ultrasound or MRI can be used to localize areas of inflammation.

Treatment of bursitis is almost always conservative, including rest, ice, NSAIDs, gentle stretching, and physical therapy. Ultrasound-guided steroid injections may help to alleviate the painful symptoms of bursitis. In recalcitrant cases, surgical bursectomy or tendon release may be helpful (open or arthroscopic).^{17,75,166,188}

Athletic Pubalgia

Athletic pubalgia has unfortunately become a term referring to pain around the groin, which can encompass osteitis pubis, adductor dysfunction, sports hernia, or other pelvic/abdominal muscular injuries. However, most clinicians regard athletic pubalgia as meaning a sports or "sportsman's" hernia,^{120,124,162} although there is no universally accepted definition. Adductor dysfunction causes tenderness localized to the adductor longus insertion; on examination there is pain with adduction against resistance and with stretch in abduction. This is usually managed nonoperatively, with measures including rest and physical therapy. Occasionally, the use of injected corticosteroids can give relief,¹⁵⁷ and adductor tenotomy has been shown to provide long-term relief in recalcitrant cases.^{4,108}

Osteitis pubis causes tenderness to palpation at the pubic symphysis. Chronic osteitis pubis can cause chronic changes that induce inappropriate osteoclastic activity and osseous resorption on plain radiographs^{112,115} as well as the "secondary cleft sign" on MRI²⁰; edema is seen around the pubis on MRI in acute

cases.^{128,175} Osteitis pubis is treated normally with rest and physical therapy,⁶⁹ with injections,^{70,139} and with surgical intervention^{130,182} in recalcitrant cases.

Athletic pubalgia is a syndrome resulting from weakness of the posterior wall of the inguinal canal causing nerve irritation and disruption and instability of the muscular attachments.^{3,102,120,132,172} Athletic pubalgia can be thought of as an overuse injury^{7,58} in which the muscular attachments of the pelvis are unbalanced, which leads to a loss of postural control and an increase in the shear forces across the pelvis. Most patients report insidious unilateral, dull, achy pain in the groin that is significantly exacerbated by activities such as running, cutting, or twisting. Coughing, sneezing, and Valsalva-type maneuvers usually worsen the pain. Physical examination can show tenderness over conjoined tendon, adductor origin, distal rectus insertion, and the inguinal canal. It is important to distinguish athletic pubalgia from an inguinal or femoral hernia. Newer MRI imaging can be useful, showing rectus abdominis tendon injury and subtle abnormalities of the myofascial layers of the abdominal wall.^{5,187} Prevention with measures to improve core and hip stability and flexibility is paramount in decreasing the incidence of athletic pubalgia.¹²¹ Nonoperative measures include rest, antiinflammatories, strengthening of the core musculature, and postural control.^{39,94} Surgical exploration should be considered if nonoperative treatment has failed after 3 to 6 months and other diagnoses have been excluded. Surgical measures include primary pelvic floor repair without mesh,^{119,131} open anterior mesh repair,^{65,81} and laparoscopic mesh repair.⁴² Surgical repairs should be referred to an appropriate surgeon.

Snapping Hip (Coxa Saltans)

A snapping hip is characterized by an audible or visible popping or snapping sound when the hip is brought through a certain ROM. It can be painful and is often exacerbated by sporting activities, particularly with running up and down a hill. Three typical types of snapping hip have been described: lateral or external (iliotibial band), internal or medial (psoas), and intra-articular.⁶ The external type is the most common and involves the posterior border of the iliotibial band or the anterior border of the gluteus maximus tendon snapping over the greater trochanter when the hip is flexed from an extended position or with internal and external rotation while the hip is extended.¹⁵⁶ The external snapping is often visible as the iliotibial band clunks over the greater trochanter. The internal snapping hip is the iliopsoas tendon snapping over the iliopectineal eminence or the femoral head; it is often painful and audible.⁴⁰ The intra-articular snapping can be caused by a loose body, chondral flap, labral tear, or synovial plica.^{8,113}

The history and physical is usually diagnostic for the location and cause of the snapping hip. Patients with external snapping hip can often reproduce the symptoms voluntarily. Internal snapping hip occurs when the hip is moved from a flexed and abducted position to an extended and adducted position; this can often be seen by ultrasonography. MRI is indicated if intra-articular pathology is suspected. Initial management of coxa saltans externa or interna includes physical therapy for stretching of the iliotibial band or iliopsoas. Antiinflammatory medications, rest, and bursal

injections can also be effective. If conservative management does not relieve the symptoms, surgical intervention may be considered. Open surgical approaches involving fractional lengthening or release have been performed,^{44,173,181} although few reports have been described in adolescents.⁴⁰ Recently arthroscopic treatment has been described with good short-term success for both coxa saltans interna and externa.^{75,76,78,80,177}

LABRAL TEARS

The acetabular labrum is a ring of fibrocartilage that almost encircles the acetabulum of the hip joint and is believed to deepen the acetabulum, enhance joint stability, and provide a fluid seal for the articulating femoral head and acetabulum.⁴⁵⁻⁴⁷ Injuries to the labrum occur during sports because of mechanical stresses or twisting in sports such as football, soccer ballet, ice hockey, and others. The patient may complain of hip or groin pain and experience catching or locking of the hip during certain maneuvers. Disorders such as hip dysplasia and femoroacetabular impingement increase the risk of labral tears. Tears can occur in any part of the labrum but are more common in the anterosuperior portion of the acetabulum, perhaps due to the higher stresses in this region.¹⁰¹ Radiographic imaging of the pelvis and hip can demonstrate a dysplasia, femoroacetabular impingement, or other abnormality, but an MR arthrogram is the most commonly used modality for diagnosing labral tears.^{23,85}

Arthroscopy has become the standard for diagnosing and treating acetabular labral tears. Conservative management may alleviate pain, but surgical repair or débridement is the most definitive treatment (Fig. 136.5). Several studies have shown good long-term results and return to sports after labral treatment in both adults and adolescents.^{26,74,89,165} When irreparable tears are encountered in the adolescent athlete, the labral sealing mechanism can be reestablished by performing a reconstruction of the labrum using graft tissue (autograft or allograft hamstring, iliotibial band, tibialis anterior, or posterior or ligamentum).⁹

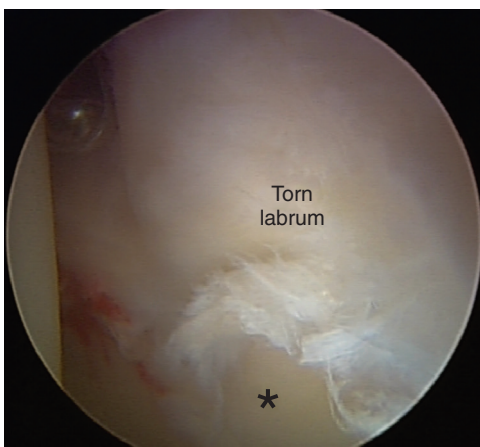


Fig. 136.5 Arthroscopic picture of a left hip depicting a large tear in the labrum. The black asterisk denotes the articular surface of the acetabulum.

ACQUIRED DISEASE

Femoroacetabular Impingement

Femoroacetabular impingement has been gaining increasing attention as a common etiology of hip pain in athletes.^{52,53} Anatomically, the concept of femoroacetabular impingement is that certain minor morphologic alterations to the hip joint combined with repetitive abutment of the femoral head-neck junction against the anterior rim of the acetabulum lead to chondrolabral dysfunction and eventual early degeneration of the hip joint. The morphologic alterations may be either on the acetabular side (pincer impingement) or the femoral head-neck side (cam impingement) or both.⁹⁵ FAI is now being recognized as a significant cause of hip pain in the adolescent population.¹⁶³

Patients often present with an insidious onset of groin pain that is made worse with sporting activity. There may be an associated labral tear and patients often describe their pain with a hand cupped over the anterolateral hip with the thumb and forefinger in the shape of a "C."²⁵ Impingement can be detected with decreased hip flexion and limited internal rotation and a positive impingement (flexion, adduction, and internal rotation) test. Radiographic imaging should include an anteroposterior (AP) view of the pelvis and orthogonal views of the hips.³⁰ A bony prominence at the head-neck junction is indicative of a cam lesion, whereas acetabular overcoverage or retroversion on the radiograph is termed a pincer. Additional MRI imaging can be helpful for evaluation of the soft tissues of the joint, including the cartilage and labrum.⁹⁹

Treatment goals include improving hip muscle flexibility, pelvic tilt, strength, and posture. Progression of arthritic changes from impingement can be gradual but if left untreated can cause severe permanent injury to the articular surfaces of the joint. Refractory pain should be managed by either open surgical dislocation or arthroscopic treatment.^{52,92} Good results have been achieved in the adolescent population with arthroscopic intervention^{33,43,148,149} even in revision cases.¹³⁵

Legg-Calve-Perthes

Legg-Calve-Perthes disease is an idiopathic self-limiting condition involving avascular necrosis of the femoral epiphysis. The exact etiology of the disease is largely unknown, although a number of vascular causes have been proposed. In the 100 years since this disease was first described there has been little insight into its etiology and pathophysiology. It typically presents within the first decade of life predominantly among males 4 to 8 years of age.⁸⁷ Bilateral involvement is uncommon, occurring about 15% of the time, but is almost never simultaneous. The pathogenesis is complex and passes through multiple stages, initially avascular necrosis, fragmentation, resorption, and collapse and then reossification. The Catterall²⁸ and Herring⁶⁷ classifications are the two most common systems used to give a prognosis depending on the location of necrosis and percent of femoral head involvement. The natural history of the disease is variable, largely dependent on age at the time of diagnosis and degree of involvement of the femoral head. Age less than 9 years and less than 50% of involved femoral head are predictive factors for a good prognosis.⁶⁷ The younger the child is at the time of the

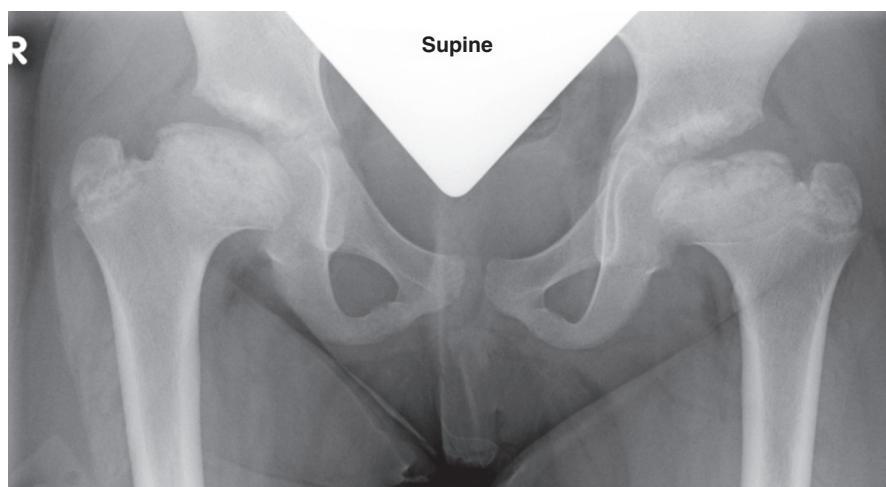


Fig. 136.6 Bilateral Legg-Calve-Perthes disease. The right side of the femoral head is in the reossification stage while the left side is in the fragmentation stage.

onset of disease, the greater the time there is for subsequent growth and remodeling. Fifty percent of patients who had childhood Perthes disease who did not receive treatment developed osteoarthritis in the fifth decade of life.¹²³

Perthes disease is specific to the hip joint and typically presents as an insidious unilateral painless limp. If pain is present it is usually mild, exacerbated by activity and commonly referred to the knee. The most common physical examination findings include decreased internal rotation and abduction of the hip. AP and lateral radiographs are diagnostic (Fig. 136.6). The goal of treatment is to try to contain the femoral head within the acetabulum, which can be performed by traction, casting, abduction bracing, tendon releases, or osteotomies of the femur or acetabulum. At present there is little conclusive information in the literature regarding one treatment method over the other. In patients with severe disease, surgical intervention appears to be preferable compared with nonoperative treatment. Patients over the age of 8 classified as B or B/C according to the Herring classification have a better outcome with surgical intervention.⁶⁷ Adolescents who have had a history of Legg-Calve-Perthes disease may present later in life with an abnormally shaped femoral head that predisposes them to FAI. Arthroscopic treatment of residual cam deformity from Perthes patients showed restoration of hip geometry and good short-term results.^{50,79}

Slipped Capital Femoral Epiphysis

Slipped capital femoral epiphysis (SCFE) involves the posterior displacement of the proximal femoral epiphysis with concomitant extension and external rotation of the femoral neck and shaft. It is a common hip disorder of adolescence, occurring most often in obese African American males between 10 and 16 years of age.¹⁴⁶ SCFE is usually associated with pubertal growth but may be due to trauma, endocrine disorders, or inflammatory disorders. Increased body mass index is also a significant risk factor.¹⁰⁴ Bilateral involvement can occur in up to 60% of cases.¹⁰³ The classification of SCFE has been traditionally based on the acuity of symptoms and severity of the slip; however, a greater emphasis is now being placed upon mechanical stability of the

slip due to a greater prognostic value. A mechanically stable slip is such that the patient can bear weight with or without crutches, whereas a patient with an unstable slip is unable to bear weight because of pain.¹⁰⁵ An unstable slip typically represents an acute physeal fracture with displacement.

Accurate, early diagnosis of SCFE is important to prevent short- and long-term complications, including chondrolysis of the joint and avascular necrosis of the femoral head. The average duration of symptoms is 5 months for stable slips. The complaints are often insidious and ambiguous, with activity related hip, thigh, or knee pain.⁹¹ The delayed onset of significant pain and dysfunction may allow for progression of a stable to an unstable slip, which has major implications on long-term prognosis.⁸⁸ Physical examination shows obligate external rotation of the affected side; AP and frog-leg lateral radiographs are diagnostic. If a slip is suspected and is negative on plain films, an MRI should be obtained (Fig. 136.7). Treatment is always operative, with the gold standard now involving in situ pinning with a single screw crossing the physis. Severe slips with significant deformity have recently been treated with a modified Dunn osteotomy through a surgical dislocation approach.^{73,100,167,186} However, there is a clear relationship between stability and severity of the slip and the postoperative risk of avascular necrosis. Additionally, a cam lesion can develop at the time of the SCFE. Although remodeling can occur,^{35,37} damage to the acetabular labrum and cartilage has been noted with both open and arthroscopic approaches.^{96,98,164} Multiple reports have detailed good results with arthroscopic resection of the resulting cam deformity from the SCFE along with screw removal.^{12,29,72,96,98,174,185} Additionally, a recent report on arthroscopic subcapital realignment osteotomy appears promising.¹⁵²

CONCLUSION

Although there are a multitude of injuries that can occur around the hip and pelvis, some occur as a result of a disease process whereas most occur as the result of trauma. The prevention of such injuries in the adolescent and pediatric population is

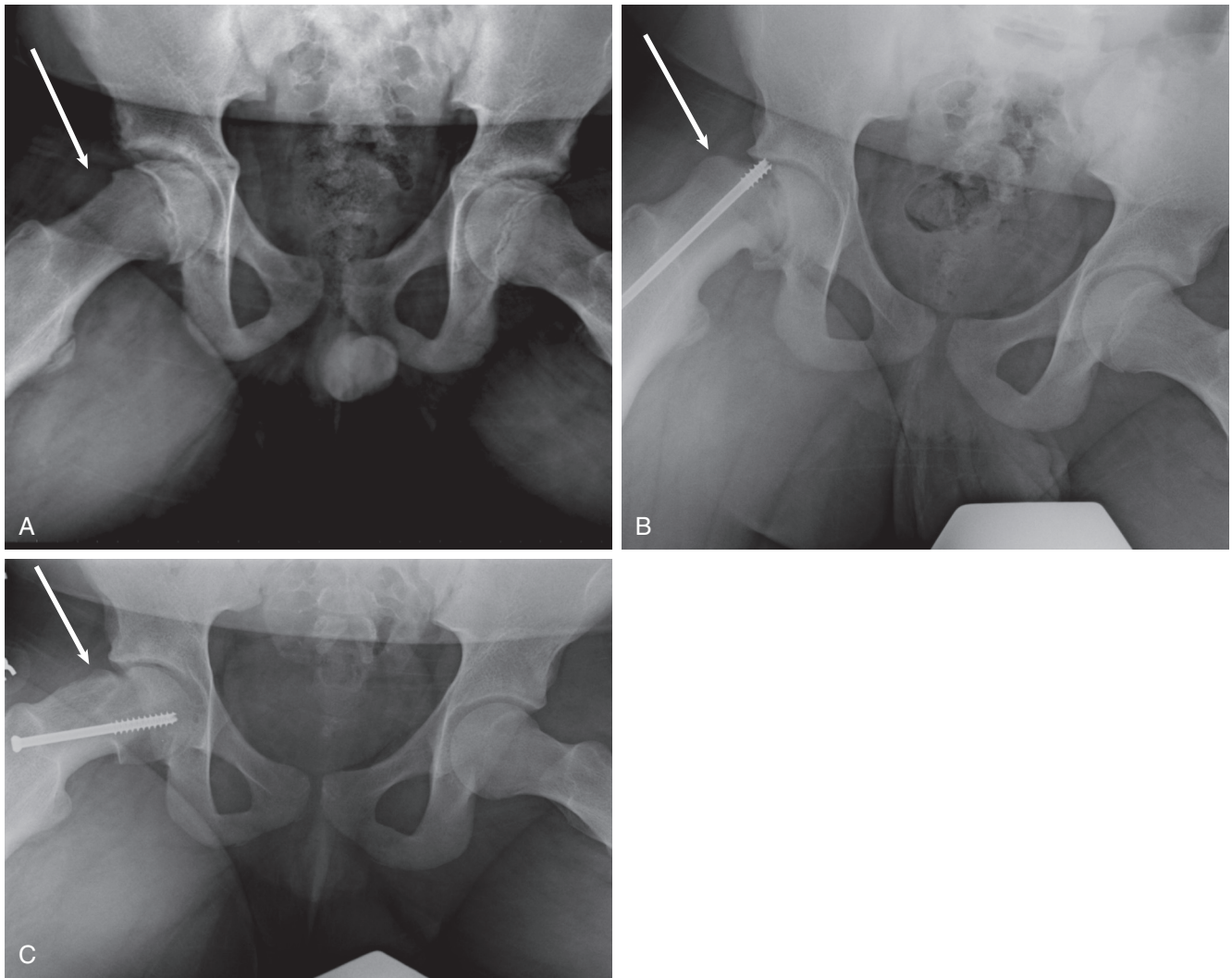


Fig. 136.7 (A) Mild slipped capital femoral epiphysis on the right hip that underwent in situ pinning with a single cannulated screw. The *white arrow* depicts the slip. (B) In the same patient the epiphysis grew off the screw and continued to slip. The *white arrow* shows the increase in the metaphyseal bump. (C) After repinning of the slipped capital femoral epiphysis with fusion of the physis. The *white arrow* shows the residual femoroacetabular impingement.

immensely important. Almost all sports involve the use of the lower extremity and core stability. A number of studies have linked the lack of core stability with the development of hip and pelvic injuries.^{97,133,142,170,183} These studies underscore the importance of developing, strengthening and maintaining the flexibility of the core muscle group. Improvement to the core muscle group can help to decrease the likelihood and/or severity of hip and pelvic injury in the adolescent athlete. Further research and programs to develop and improve strategies to prevent injury are warranted.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Eberbach H, Hohloch L, Feucht MJ, et al. Operative versus conservative treatment of apophyseal avulsion fractures of the

pelvis in the adolescents: a systematical review with meta-analysis of clinical outcome and return to sports. *BMC Musculoskelet Disord.* 2017;18(1):162.

Level of Evidence:

V

Summary:

This meta-analysis of 596 patients with avulsion fractures of the pelvis provides evidence for operative versus nonoperative treatment.

Citation:

Ganz R, Parvizi J, Beck M, et al. Femoroacetabular impingement: a cause for osteoarthritis of the hip. *Clin Orthop Relat Res.* 2003;417:112–120.

Level of Evidence:

IV

Summary:

This article is the sentinel paper on femoroacetabular impingement.

It has significantly influenced the understanding of hip pathology and subsequent treatment regimens.

Citation:

Millis MB, Lewis CL, Schoenecker PL, et al. Legg-Calve-Perthes disease and slipped capital femoral epiphysis: major developmental causes of femoroacetabular impingement. *J Am Acad Orthop Surg*. 2013;21(suppl 1):S59–S63.

Level of Evidence:

V

Summary:

An excellent review of childhood hip disorders causing femoroacetabular impingement. this article underscores the development of irreversible chondral damage and advocates early treatment.

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